

Part 1: Summary of Lecture 3 (Arithmetic & Logic)

1. Shift Operations & Mathematical Effect:

- **Logical Left Shift (LSL):** Each shift **multiplies** the number by 2.
- **Logical Right Shift (LSR):** Each shift **divides** the number by 2 (reduces it by half).
- **Cyclic Shift (Rotate):** Bits shifted out from one end re-enter from the other end. Can be done **Through Carry** or directly.

2. Signed Multiplication (2's Complement):

- **Rule:** You **must sign-extend** the operands to the full product bit width before multiplying.
 - *Example:* To multiply 4-bit numbers, extend them to 8 bits first.
- **Without Extension:** Multiplying negative numbers directly (e.g., 101 for -3) leads to an **error** in the result.
- **Hand Calculation:** It is often easier to multiply the positive values and then adjust the sign (2's complement) at the end.

3. Division:

- Performed using a "Shift and Subtract" method similar to long division.
- If the divisor can be subtracted from the current bits put **1** in quotient. If not put **0**.

4. Condition Codes (Flags - NVZC):

- **C (Carry):** Holds the bit that was last shifted out.
- **V (Overflow):** Indicates if the result size exceeds the register capacity (sign change error).
- **Example from Slides:** Shifting 0111 Left Result 1110.
 - **C=0** (The MSB 0 was shifted out).
 - **V=0** (No overflow occurred).